INCH-POUND
MIL-DTL-11040H
31 May 2022
SUPERSEDING
MIL-DTL-11040G
w/AMENDMENT 2
13 September 2017

DETAIL SPECIFICATION

BELT, V: ENGINE ACCESSORY DRIVE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for standard and heavy duty endless V-type engine accessory drive belts of various lengths and belt widths. The belts are either plain or notched or cogged base.

1.2 Classification. Belts are of the following grades, styles and belt widths. (See 1.2.1, 1.2.2, and 1.2.3)

1.2.1 Grade. The V-belts are of the following grades.
A - Standard duty. See 3.3.1.1
B - Heavy duty. See 3.3.1.2

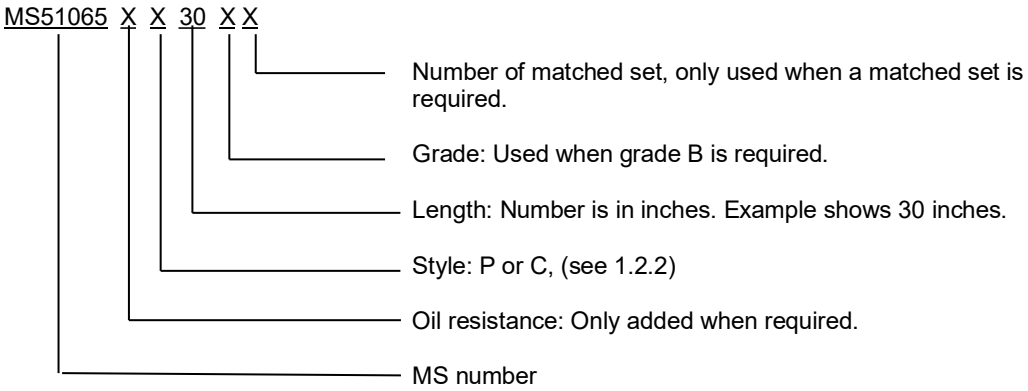
1.2.2 Style. The V-belts are of the following styles.
P - Plain base.
C - Notched or cogged base.

1.2.3 Belt width. The V-belts are of the following widths and belt thickness:

0.380 (9.65) X .312 (7.92) MS51065	0.750 (19.05) inch X 0.438 (11.13) MS51068
0.500 (12.70) X .406 (10.31) MS51066	0.875 (22.23) inch X 0.500 (12.70) MS51069
0.688 (17.48) X .406 (10.31) MS51067	1.000 (25.40) inch X 0.563 (14.29) MS51070

1.2.4 Part or Identifying Number (PIN).
For a complete PIN description of the specific PIN refer to the required specification sheet.

PIN example:



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property (See supplement 1 for list of specification sheets.)

(Copies of these documents are available online at https://assist.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

THE ASSOCIATION FOR RUBBER PRODUCTS MANUFACTURERS

ARPM IP-3-4 - Power Transmission Belt Technical Bulletin

(Copies of these documents are available online at https://arpminc.com/.)

ASTM INTERNATIONAL

ASTM D471 - Standard Test Method for Rubber Property, Effect of Liquids

(Copies of these documents are available online at https://astm.org.)

ISO - INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO/IEC 17025 - General requirements for the competence of testing and calibration laboratories

(Copies of these documents are available online at https://www.iso.org.)

SAE INTERNATIONAL

SAE J636 - V-Belts and Pulleys

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Belts furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.4 and 6.3).

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3.2.1 Qualification by similarity. With written approval from the qualifying activity, qualification of belts of 0.380 inch top width will qualify a manufacturer, at his request, for all belts of top width of 0.500 inch or less and qualification of belts of 0.688 inch top width will quality a manufacturer, at his request, for all belts of top width greater than 0.500 inch provided the belts have the same design and manufacturing process as specified herein.

3.3 Materials. Materials shall be as identified herein or as approved by the qualifying activity. However, when a definite material is not specified, a material shall be used which will enable the belts to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of acceptance of the finished product.

3.3.1 Cords. Cords shall be as specified herein, shall be pre-stretched, and shall be impregnated with a rubber compound.

3.3.1.1 Grade A. Cords for grade A belts shall be natural or synthetic.

3.3.1.2 Grade B. Cords for grade B belts shall be polyester. (Grade B with polyester cords can replace Grade A if approved by the program office)

3.3.2 Fabric. Fabric shall be woven or knit of natural or synthetic fibers, and shall be impregnated with rubber.

3.3.3 Rubber. Rubber shall be either natural or synthetic.

3.3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction. Each belt shall be an endless loop having a trapezoidal cross-section with a belt width and a belt thickness as specified in table I. The belt shall consist of tension members (inserts) and a compression member. Covers may be used. The members shall be tightly banded with rubber.

3.4.1 Tension members. Tension members shall be uniformly spaced in a direction parallel to the circumference of the belt, and shall be imbedded in the belt structure.

3.4.2 Compression members. Compression members shall form the narrower trapezoidal portion of the belt in the section below the tension members and shall support the belt in the pulleys. The base that forms the inner circumference of the belt shall be plain, notched, or cogged.

3.4.3 Cover. The cover, if used, shall consist of one or more plies of bias-cut fabric or cord, frictioned or impregnated with rubber.

3.4.4 Splicing. Not more than two transverse splices shall be allowed in any ply or fabric within a length of 66 inches or less.

3.4.5 Sizes. Sizes of belts (widths and effective lengths) shall be as specified herein and as specified in, MS51065, MS51066, MS51067, MS51068, MS51069, and MS51070 or as specified in table I (see 6.2).

TABLE I. Belt cross-section. 1/

Document standard	Belt width		Belt thickness	
	Inches	mm	Inches	mm
MS51065	0.380	09.65	0.313	07.95
MS51066	0.500	12.70	0.406	10.31
MS51067	0.688	17.48	0.406	10.31
MS51068	0.750	19.05	0.438	11.13
MS51069	0.875	22.23	0.500	12.70
MS51070	1.000	25.40	0.562	14.27

1/ Dimensions are in inches, metric equivalent are given for information only.

3.4.6 Oil resistance. When specified (see 6.2), the belt shall be oil resistant.

3.5 Belt matched sets. When specified (see 6.2), belts shall be furnished in matched sets of the size and number of belts required.

3.6 Performance.

3.6.1 Examination of product. The belt shall conform to the requirements of this specification and the applicable specification sheet when visually examined as specified in 4.7.1.

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3.6.2 Effective length tolerance.

3.6.2.1 Single belts. The effective length tolerance for grade A single belts, when tested as specified in 4.5.2, shall not exceed the center distance tolerance specified in table II. The effective length tolerance for grade B single belts when tested as specified in 4.7.2 shall not exceed the center distance tolerance specified in table III.

TABLE II. Effective length tolerance, grade A.

Length	Effective belt length (inches)	Tolerance on center distance 1/			
		Plus		Minus	
		Inches	mm	Inches	mm
L1	40 and less	0.125	3.18	0.156	3.96
L2	Over 40 and through 50	0.125	3.18	0.188	4.78
L3	Over 50 and through 60	0.156	3.96	0.219	5.56
L4	Over 60 and through 80	0.188	4.78	0.281	7.14
L5	Over 80	0.219	5.56	0.344	8.74

1/ Dimensions are in inches, metric equivalent are given for information only.

TABLE III. Effective length tolerance, grade B.

Effective belt length (inches)	Tolerances on center distance plus or minus 1/	
	Inches	mm
50 and less	0.125	3.18
Over 50 and through 60	0.156	3.96
Over 60 and through 80	0.188	4.78
Over 80	0.219	5.56

1/ Dimensions are in inches, metric equivalent are given for information only.

3.6.2.2 Matched set. Belts of matched sets shall conform to the effective-length tolerance specified in table II for grade A belts and table III for grade B belts, except that the difference in effective length, between test pulley centers of the longest and shortest belt in any one matched set, shall not exceed .062 inch for grade A belts, and the differences in length specified in table IV for grade B belts, when tested as specified in 4.7.2.

TABLE IV. Matched set tolerance, grade B belts. 1/

Belt width		Difference in length between test pulley centers	
Inches	mm	Inches	mm
0.380	9.65	0.04	1.02
0.500	12.70	0.04	1.02
0.688	17.48	0.06	1.52
0.750	19.05	0.06	1.52
0.875	22.23	0.06	1.52
1.000	25.40	0.06	1.52

1/ Dimensions are in inches, metric equivalent are given for information only.

3.6.3 Rideout. The top of the belt with respect to the top of the pulley groove shall be within the limits specified in table V, when tested as specified in 4.7.3 (see figure 2).

TABLE V. Rideout. 1/

Belt width		Rideout ± 0.031 (0.79 mm)	
Inches	mm	Inches	mm
0.380	9.65	0.063	1.60
0.500	12.70	0.063	1.60
0.688	17.48	0.094	2.39
0.750	19.05	0.094	2.39
0.875	22.23	0.094	2.39
1.000	25.40	0.094	2.39

1/ Dimensions are in inches, metric equivalent are given for information only.

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3.6.4 Electrical resistivity. The electrical resistivity of the belt shall be not greater than 300,000 ohms when tested as specified in 4.7.4.

3.6.5 Oil-resistance. When specified (see 6.2) the volume swell shall increase no more than 30 percent after exposure to IRM 903 oil (see ASTM D471), and shall have a volume swell of no more than 15 percent after exposure to IRM 901 oil (see ASTM D471), when tested as specified in 4.7.5.

3.6.6 Resistance to temperature. The belts shall not crack, break, or require more torque to start or continue rotation than that specified in table VI, when tested as specified in 4.7.6.

TABLE VI. Maximum torque requirements.

Top width (inches)	Torque required to start (maximum)	Torque required to continue (maximum)
Less than 0.688	20 ft. lbs.	12.5 ft. lbs.
0.688 through 0.750	25 ft. lbs.	15.0 ft. lbs.
0.875 through 1.000	30 ft. lbs.	20.0 ft. lbs.

3.6.7 Resistance to fatigue. The belts shall not crack, break, or slip in excess of 8 percent after two permissible adjustments to compensate for slippage when tested as specified in 4.7.7.

3.7 Age. Age limits shall be as per storage guidelines in ARPM Bulletin IP-3-4 Storage of Power Transmission Belts. The quality of belts has not been found to change significantly within seven years of proper storage at temperatures less than 85° F (30° C) and relative humidity below 70 percent. Also, there must be no exposure to direct sunlight. When specified, the age of the belts shall be not more than 12 months (4 quarters) old from the time of manufacture to the date of acceptance by the Government.

3.8 Marking.

3.8.1 Single belts. Unless otherwise specified, the belts shall be marked on the outside circumference in accordance with MIL-STD-130, items identified by Military or Industry Association Numbers; in addition, the date of manufacture shall be shown by quarter and year (example 1Q98), or month and year (example Jan 98) or week and year (example 3-98). Ink imprinting or permanent mold marking is acceptable. An alternative marking is shown in Figure 1 below.

PRODUCTION DATE CODE GENERATION (No "I" or "O")	
FORMAT: MONTH, YEAR, DAY	CODE
EX: August 20, 2021	HB20
EX: August 20, 2044	HAA20

MONTH LIST	CODE
JANUARY	A
FEBRUARY	B
MARCH	C
APRIL	D
MAY	E
JUNE	F
JULY	G
AUGUST	H
SEPTEMBER	J
OCTOBER	K
NOVEMBER	L
DECEMBER	M

YEAR LIST	CODE
2020	A
2021	B
2022	C
2023	D
2024	E
2025	F
2026	G
2027	H
2028	J
2029	K
2030	L
2031	M
2032	N
2033	P

YEAR LIST	CODE
2034	Q
2035	R
2036	S
2037	T
2038	U
2039	V
2040	W
2041	X
2042	Y
2043	Z
2044	AA
2045	BB
2046	CC
2047	DD

Figure 1. Alternative Marking for Single Belts

3.8.2 Matched sets. Each belt of a matched set shall be marked as specified in 3.8.1. In addition, all belts of a matched set shall be tied together. A tag or separate label shall include the following notice: "WARNING: THIS IS A MATCHED SET. DO NOT BREAK TIES EXCEPT AT INSTALLATION."

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3.9 Workmanship. The belt shall be fully formed and symmetrical. When laid on a flat surface, the belt shall exhibit no evidence of twisting. The belt shall exhibit no evidence of delamination, chatter marks or melting. The outer circumference of the belt shall be square with the inner circumference. The belt shall be free from imbedded foreign material and shall be clean on the exterior.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Conformance inspection (see 4.5).
   1. Individual tests (see 4.5.2).
   2. Sampling tests (see 4.5.3).

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment (i.e. Industry Standard, Military Standard, etc.) shall be in accordance with ISO/IEC 17025 or equivalent.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the qualifying activity on sample units produced with equipment and procedures actually used in production. Qualification inspection shall be as specified in table VII.

TABLE VII. Inspection requirements.

Inspection or test	Requirement paragraph	Test	Qualification	Quality conformance inspection	
				Individual tests 1/	Sampling tests
Examination of product	3.6.1	4.7.1	X	X	
Effective length tolerance	3.6.2	4.7.2	X		X
Rideout	3.6.3	4.7.3	X		X
Electrical resistivity 2/	3.6.4	4.7.4	X		
Oil resistance 2/	3.6.5	4.7.5	X		
Resistance to temperature 2/	3.6.6	4.7.6	X		
Resistance to fatigue 2/	3.6.7	4.7.7	X		

1/ 100% inspection required on all V-belts supplied to this specification. 2/ These are destructive test.

4.4.1 Samples for qualification. Samples for qualification shall be representative of the products proposed to be furnished to the Government. Samples shall consist of three belts of each grade and belt width manufactured by the same process.

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of individual tests and sampling tests.

4.5.2 Individual tests. Each belt shall have been subjected to and passed all the individual tests specified in table VII. Any belt which fails any test criteria in the individual test sequence shall be removed from the lot at the time of observation or immediately at the conclusion of the test in which the failure was observed.

4.5.3 Sampling tests. Sampling tests shall consist of the tests specified in table VII in the order shown. These tests shall be performed on a production lot basis. Random samples shall be selected to form an inspection sample. If one or more defects are found in the inspection sample, then the production lot shall be inspected for that particular defect and defects removed. A second inspection sample shall be selected from the production lot and all sampling tests again performed. If one or more defective items are found in the second inspection sample, the production lot shall be rejected and shall not be supplied to this specification.

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4.5.3.1 Production lot. A production lot shall consist of belts manufactured on the same production line(s) by means of the same production techniques, materials, controls, and design during the same production run.

4.5.3.2 Inspection sample. The inspection sample shall be product selected at random from the production lot without regard to quality and shall be the size specified in table VIII.

TABLE VIII. Inspection sample.

Production lot Size	Accept on Zero sample Size
1 to 8	All
9 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1,200	27
1,201 to 3,200	35
3,201 to 10,000	38
10,001 to 35,000	46

4.5.4.1 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4.2 Retention of qualification. To retain qualification, the contractor shall forward a report at 36-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

A summary of the results of the sampling tests, indicating as a minimum the number of lots and quantities that have passed and the number that have failed. The results of tests of all reworked sampling lots shall be identified and accounted for.

4.5.4.3 Periodic qualification report. Failure to submit the report within 30 days after the end of each ~~24~~36-month period may result in loss of qualification for the product. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements.

4.5.4.4 Changes and notification. The manufacturer shall be responsible for the implementation of any major changes of the product or quality assurance program which may affect performance, quality, and reliability. Notification to the qualifying activity (QA) and acquiring activity of major changes that may affect the performance, quality, reliability, or interchangeability to the product or information as listed in the certification/laboratory suitability received from the Qualifying Activity (QA) is required prior to implementation of the changes. The manufacturer may make notification of this change of product through the GIDEP using the Product Change Notice; in any case the manufacturer shall assure that all known acquiring activities for this product and the QA are notified. Initial documentation and subsequent major changes in design, materials, or processing shall be documented and retained for a minimum of 10 years.

4.5.4.5 Discontinuation of products. The manufacturer shall have a system that provides notification of product discontinuation which comprises date of discontinuation and last time buy opportunity in advance (at least 6 months in advance) to the DLA Land and Maritime for purchasing devices.

4.5.4.6 Certificate of conformance (C of C). Manufacturers or suppliers including distributors who offer belts described by this specification shall provide written certification, signed by the corporate officer who has management responsibility for the production of the belts, (1) that the belts being supplied have been manufactured at approved facilities listed on the certification letter issued by the QA and meet the performance defined in the specification whether or not the actual testing has been performed, (2) that all belts are as described on the certificate of conformance which accompanies the shipment. The responsible corporate official may, by documented authorization, designate other responsible individuals to sign the certificate of conformance, but the responsibility for conformity with the facts shall rest with the responsible corporate officer. The certification shall be confirmed by documentation to the Government or to users with Government contractors or subcontractors, regardless of whether the belts are acquired directly from the manufacturer or from another source such as a distributor. When other sources are involved, their acquisition certification shall be in addition to the certificates of conformance and acquisition traceability provided by the manufacturer and previous distributors.

4.5.4.7 Records Retention. All records required by this document shall be retained for a minimum of 10 years unless otherwise stipulated in the Department of Defense specification applicable to the parts being tested or extended by customer requirements.

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4.6 Test conditions.

4.6.1 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure.

4.7 Methods of inspection. The following identified tests and test methods assure belt integrity within typical operating conditions and applications. Alternate commercial industry standard test methods are allowed; however when an alternate method is used, documented approval must be obtained from the qualifying activity prior to the performance of the test. The test methods described herein are proven methods and shall be the referee method in case of dispute.

4.7.1 Examination of product. The belt shall be visually examined for the defects specified in table IX, defects for visual examination. With documented approval from the qualifying activity, statistical quality control may be used for visual examination. Requirements shall be as specified in 3.6.1.

TABLE IX. Defects for visual examination.

Defects	Defects
101. Design of belt not as specified 102. Tension members frayed, knotted, or not completely imbedded. 103. Compression member not of symmetrical profile or contains cavities or melted spots 104. Age of belt over 7 years or 12 months, when specified or supplied to customer 105. Indentation, any side, top or bottom 106. Press lap defect 107. Flow distortion 108. Flash untrimmed (acceptable if removed) 109. Protrusion, wearing side 110. Blister, any side 111. Foreign material, wearing side 112. Open longitudinal seam, any side 113. Loose transverse seam, wearing side 114. Torn cover, wearing side 115. Trim cut 116. Cogging cut-out skipped 117. Marking, with wrong information 118. Marking, missing or illegible 119. Matching faulty 120. Porosity in cogging area 121. Scratch through ply	201. Protrusion, top or bottom 202. Foreign material, top or bottom 203. Open longitudinal seam, top or bottom 204. Loose transverse seam, top or bottom 205. Torn cover, top or bottom 206. Identification marking not in accordance with MIL-STD-130 207. Matched sets not tied together with warning notice 208. Scratch, not through a ply 209. Cogging missed (one cog) 210. Wrinkle in cover, tension member not distorted 211. Cogging cut-out overlap 212. Mold induced ridges on top of belts

4.7.2 Effective length tolerance. The effective length tolerance shall be determined by belt length. The belt shall be subjected to the length test specified in SAE J636. Conformance to effective length tolerances for grade A belts, grade B belts and matched sets shall be as specified in 3.6.2.

4.7.3 Rideout. The rideout of the belt shall be determined in accordance with the rideout test specified in SAE J636. Conformance shall be as specified in 3.6.3 (see figure 2).

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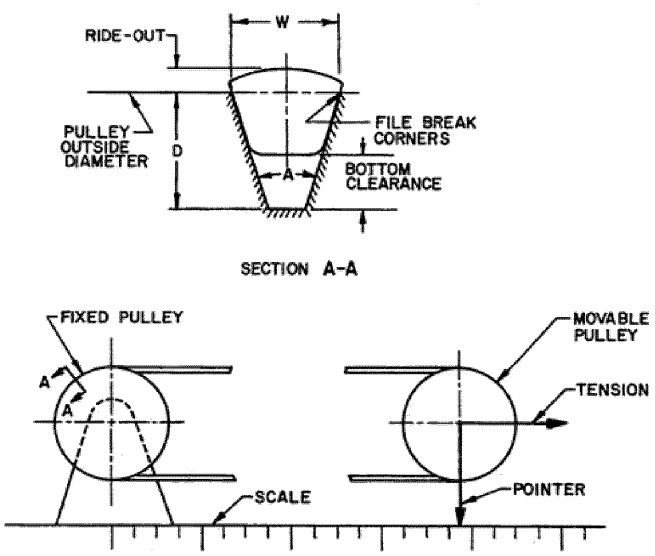
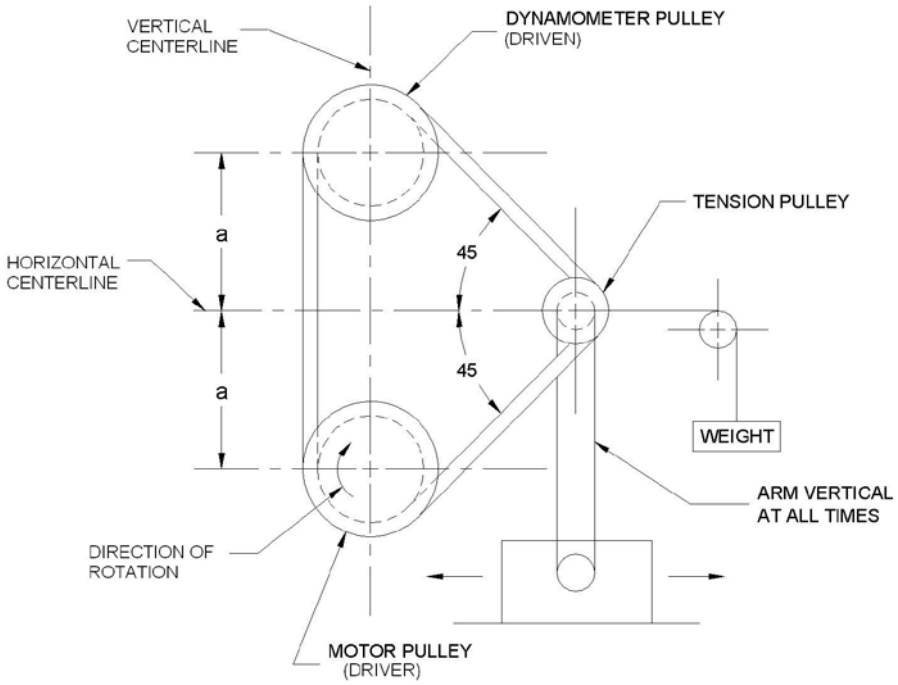


FIGURE 2. Test fixture and pulleys for determining belt effective length, rideout and bottom clearance.



NOTE: Dimension “a” is adjusted for various length belts to maintain tension pulley midway vertical centering between driver and driven pulleys.

FIGURE 3. General pulley arrangement for low temperature and fatigue tests.

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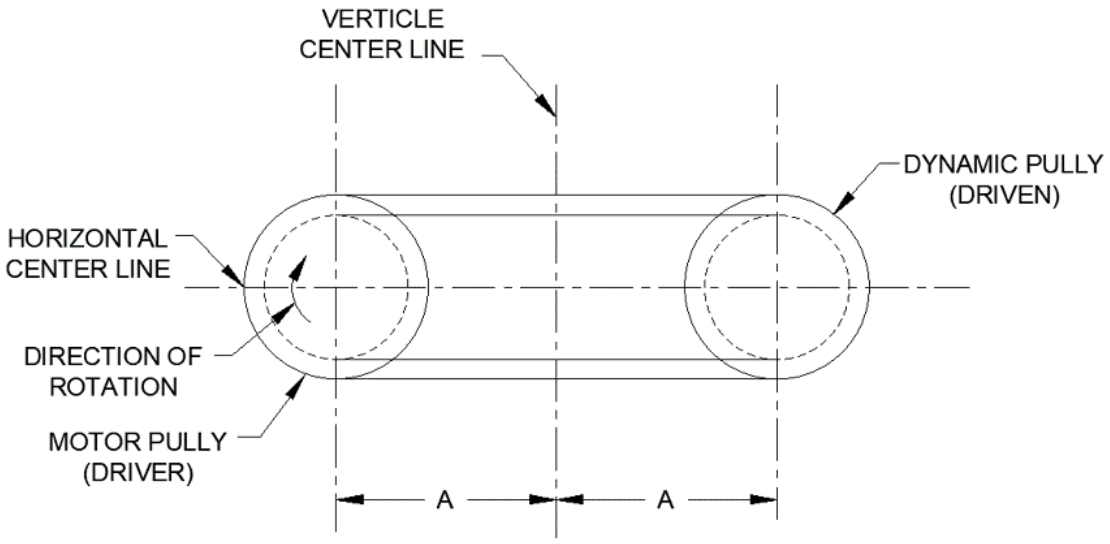


FIGURE 4. Alternative General pulley arrangement for low temperature tests.

4.7.4 Electrical resistance. This test shall be conducted as per ISO 1813

4.7.5 Oil resistance. Six belt sections, three inches long, shall be used for oil resistance test. Measure and record each belt section % volume swell at the center and .500 inch +/- .005 from each end. Three belt sections shall be immersed in IRM 901 oil and three belt sections shall be immersed in IRM 903 Oil as specified in ASTM D471. The temperature of the oils shall be maintained at a minimum of 150 °F, +/- 2 °F with the belt sections immersed in the oil for a minimum of 22 hours. At the completion of the immersion the % volume swell shall be measured. The percent increase or decrease in thickness of any specimen shall be recorded as the average of the three readings for that specimen. Conformance shall be as specified in 3.6.5.

4.7.6 Resistance to temperature. Place the belt in an air oven in an unrestrained horizontal position and maintain the temperature of the oven at 150° F, +/- 2° F, for 12 hours. At the end of 12 hours, remove the belt from the oven and allow to cool for a minimum of 30 minutes at room temperature. Then install the belt on the pulley and panel assembly (see figure 3 and in table X) and apply belt tension with the applicable weight for the width being tested as specified in table X. Alternative general pulley arrangement for low temperature tests is shown in figure 4. The tension weights may be applied as shown on figure 3 or the panel may be turned 90 degrees +/- 2 degrees with the tension pulley at the bottom and the tension weights suspended from the tension pulley. Lock tension pulley location and remove the tension weights from the belt. Then place the assembly of belts and pulleys in the cold chamber and maintain the chamber at a temperature of minus 31° F, +0/- 5° F, for 12 hours. At the end of a minimum 12 hours and while in the cold chamber, rotate the pulleys by means of a torque wrench applied to the drive pulley until the belt has been rotated not less than two complete revolutions. Record the torque to start and the torque to continue rotations. Examine the belt rotation. Evidence of cracks, breakage, or torque higher than the applicable value listed in table IV shall constitute failure of this test. Conformance shall be as specified in 3.6.6.

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TABLE X. Dimensions of pulleys for temperature and fatigue tests. 1/ 2/

Belt width		Diameter of pulley where specified groove width (Without width tolerance) occurs				Groove				Angle	
		Driver and driven pulleys ±0.010 (.254)		Tension pulley ±0.010 (.254)		Width ±0.010 (.254)		Depth min		Driver and driven pulleys ±30 min	Tension pulley ±30 min
Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	degrees	degrees
0.380	9.65	4.75	120.7	2.75	69.9	0.380	9.65	0.438	11.13	36	36
0.500	12.70	5.00	127.0	3.50	88.9	0.500	12.70	0.562	14.27	36	36
0.688	17.48	4.66	118.4	3.16	80.3	0.597	15.16	0.562	14.27	36	34
0.750	19.05	4.91	124.7	3.29	83.6	0.660	16.76	0.625	15.88	36	34
0.875	22.23	5.91	150.1	3.91	99.3	0.785	19.93	0.688	17.48	36	34
1.000	25.40	6.91	175.5	4.54	115.3	0.910	23.10	0.813	20.65	36	34

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

4.7.7 Fatigue.

4.7.7.1 Test fixture. The test fixture shall consist of three pulleys arranged as specified on figure 3 and the following:

a. A torque-reaction-type dynamometer.
b. A constant-speed motor.
c. A means for setting the tension at the value specified for the specific belt width.
d. A means of determining belt slip within an accuracy of plus or minus 1 percent.
e. A governor arranged to automatically stop the motor drive when the speed of the driven pulley (on dynamometer) drops 8 percent below that of the driver pulley (on motor).
f. Pulleys for the belt widths designated shall have dimensions as specified in table X.
g. Dial indicator.

4.7.7.2 Test procedure. Install the belt on the test fixture (see figure 3) and apply the applicable tension specified in table XI to the idler pulley. If tension is to be applied to the driven pulley, the tension values specified in table XI shall be multiplied by 1.21. Condition the belt by running with the tensile pulley unrestrained for 5 minutes, +/- 15 seconds, under the conditions specified in table XI, do not apply the dynamometer load. Stop the machine and allow the belt to condition for a minimum of 10 minutes. Turn the drive manually for 2 or 3 revolutions of the belt and, by means of a dial indicator mounted in contact with the tension-pulley support bracket, record the maximum limits of travel. Lock the tensioned-pulley support bracket in the position midway between the two limits of travel and remove the tensioning weights. Restart the machine, bring the drive up to test speed and apply the dynamometer load specified in table XI. Continue to operate the machine for 100 ±1/2 hours, in an ambient temperature of between 70°F to 90°F. If slippage exceeds 8 percent, stop the machine, let the belt stand for a minimum of 20 minutes, unlock the tensioning pulley, restore the initial tension, determine, the center position, and relock the tensioning pulley. Restart the machine, bring up to speed, reapply the dynamometer load, and continue the minimum of 100-hour run. Conformance shall be as specified in 3.6.7.

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TABLE XI. Condition of fatigue test. 1/ 2/

Belt width (inches)	Pulley diameters (inches)		Driver (rpm) minimum	Tension weight at no load (lbs) minimum	Horse-power load minimum	Belt effective length (in)
	Driver and driven ±0.010 (.254)	Idler ±0.010 (.254)				
0.380 (9.65)	4.75	2.75	4900	60	6.0	Under 40.0
	4.75	2.75	4900	70	7.0	40.0-55.0
	4.75	2.75	4900	80	8.0	55.0 & over
0.500 (12.70)	5.00	3.50	4700	80	8.0	Under 40.0
	5.00	3.50	4700	90	9.0	40.0-55.0
	5.00	3.50	4700	100	10.0	55.0 & over
0.688 (17.48)	4.66	3.16	4900	50	5.0	Under 40.0
	4.66	3.16	4900	60	6.0	40.0-55.0
	4.66	3.16	4900	70	7.0	55.0 & over
0.750 (19.05)	4.91	3.29	4700	55	5.5	Under 40.0
	4.91	3.29	4700	65	6.5	40.0-55.0
	4.91	3.29	4700	75	7.5	55.0 & over
0.875 (22.23)	4.91	3.91	3900	70	7.0	Under 40.0
	4.91	3.91	3900	80	8.0	40.0-55.0
	4.91	3.91	3900	90	9.0	55.0 & over
1.000 (25.40)	6.91	4.54	3350	80	8.0	Under 40.0
	6.91	4.54	3350	90	9.0	40.0-55.0
	6.91	4.54	3350	100	10.0	55.0 & over

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirement shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The belts covered by this specification are intended to be used on power-transmission applications that require high speed, small-pulley, short-center drives, such as drives on internal combustion engines that transmit power to generators, pumps, fans, and other accessory equipment. This specification does not apply to industrial belts. belts are intended for use in ambient temperatures as low as minus 31° F (minus 35° C)

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable QPL and Government designation.
c. Whether grade B belt is required (see 1.2). Unless otherwise specified, grade A belts should be supplied.
d. Width and effective length of belts required (see 3.4.5). Express length for belts as “effective length” in acquisition documents.
e. Whether matched sets are to be furnished and quantity of belts per set required (see 3.5).
f. Whether oil resistance is required (see 3.4.6 and 3.6.5).
g. Whether the age of belts is not to exceed 12 months (see 3.7).
h. Packaging requirements (see 5.1).
i. Country of origin.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.11040 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from (insert name, mailing address, and email of qualifying activity). An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Belts procured as components of end items. When belts covered by this specification are components of end items being procured by the Government, the requirements of ASPR 1-1107.2(b) will apply and the contracting officer should include the requirements of ASPR 1-1107.2(b) in the solicitation.

6.4 Subject term (keyword) listing.

Combustion Fans
Generators
Power-Transmission
Pulley
Pumps

6.5 Shelf life recommendation. Shelf Life Shall be as per ARPM IP-3-4. The quality of belts has not been found to change significantly within seven years of proper storage at temperatures less than 85° F (30° C) and relative humidity below 70 percent. Also, there must be no exposure to direct sunlight.

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy - YD
Air Force – 71
DLA - CC

Preparing activity:
DLA - CC

(Project 3030-2022-001)

Review activities:
Army - AR, MI, SM
Navy - MC, SH, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil

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